

Design Decisions and Trade-offs

Project 1: EU AI Act Compliance Advisor. Retrieval-augmented generation on Azure.

Cindy Moncada | AI Solutions Architect

Each entry records a decision made while building the reference implementation, the reasoning behind it, and the alternative that was considered and rejected.

1. Retrieval architecture: Foundry IQ knowledge base on an Azure AI Search vector index

Decision: Ground the agent using a Foundry IQ knowledge base backed by an Azure AI Search vector index, rather than a direct file upload or the older guided data flow.

Rationale: This is the current, supported RAG pattern on Azure. The vector index gives semantic retrieval over the Act, and Foundry IQ is the managed knowledge layer agents are meant to consume. The retrieval index itself is generally available; portal access to the Foundry IQ agentic-retrieval layer is currently preview, which I used knowingly and would note in any production plan.

Rejected: The Azure OpenAI "on your data" classic flow, which is deprecated and scheduled for retirement, so building on it would have shipped a portfolio piece on a sunset feature. Also rejected the agent's built-in file upload, which creates a separate hidden store and discards the search index and its controls.

2. Region: Sweden Central

Decision: All components (Foundry project, chat and embedding models, Azure AI Search, storage) deployed in Sweden Central.

Rationale: The advisor's subject is the EU AI Act, so EU data residency is a requirement, not a preference. Sweden Central and Spain Central both sit inside the Azure EU Data Boundary, so on residency they were equivalent. I chose Sweden Central for its broad model availability and, unlike Spain Central, its current support for the evaluation and red-teaming features later projects depend on, letting one region carry the whole plan. Region is fixed at project creation, which raised the cost of choosing wrong.

Rejected: Spain Central. Physically closest and marginally lower latency, but it lacked the evaluation and red-teaming feature support, and the latency gain was negligible for a single-user question-and-answer tool.

3. Generation model: gpt-5-mini

Decision: Use gpt-5-mini, a mini-tier model, as the answering model.

Rationale: The model's job here is to read a handful of retrieved passages and compose a grounded answer, which does not require a top reasoning tier. A mini tier delivers that at meaningfully lower cost and good latency.

Rejected: A larger reasoning tier, on cost, since the grounded-retrieval task did not justify it. Also rejected gpt-5.4-mini, my first choice, once I found it does not offer an EU Data Zone Standard deployment; keeping the residency story intact mattered more than using the newest mini.

4. Deployment type: Data Zone Standard (EU), with a documented interim

Decision: Target Data Zone Standard (EU) as the deployment type, and run on Global Standard as an interim for the reference build.

Rationale: Data Zone Standard keeps prompt and response processing inside the EU, which is the deployment type that matches an EU AI Act tool. Global Standard, by contrast, can route processing to any region where the model is available.

Rejected: Global Standard as the permanent choice, because it weakens EU processing residency. I hit a quota constraint for the EU Data Zone allocation, filed a quota request, and documented that the reference implementation runs on Global Standard in the meantime while production would require the EU Data Zone deployment.

5. Embedding model: text-embedding-3-small

Decision: Vectorize the corpus with text-embedding-3-small.

Rationale: For a single regulation of this size, the small model gives ample retrieval accuracy at lower cost per token than the large model.

Rejected: text-embedding-3-large, on cost, as its added capacity brought no meaningful retrieval benefit at this corpus size.

6. Ingestion and chunking: managed integrated vectorization

Decision: Load the corpus with the managed import and vectorize wizard, which chunked the Act into 361 retrievable passages and embedded each one.

Rationale: The managed pipeline (data source, then skillset, then indexer, then index) handles cracking, chunking, and embedding without custom code, which fits the low-code intent. Chunking matters because retrieval returns the specific relevant passages, not whole documents, which is what enables article-level citations.

Rejected: A hand-built chunking and embedding pipeline in code, which would have added engineering effort with no benefit at this scope.

7. Corpus: official consolidated text from EUR-Lex

Decision: Use the consolidated text of Regulation (EU) 2024/1689 from EUR-Lex as the single source of truth.

Rationale: A compliance tool is only as trustworthy as the document beneath it. Grounding on the official regulation makes the answers defensible and the citations real.

Rejected: Third-party summaries or explainer articles, which are unreliable as a legal source of truth and would undermine credibility in exactly the regulated-industry screening this project targets.

Note on scope

The managed-identity authentication, the least-privilege role assignments (Storage Blob Data Reader for ingestion, Search Index Data Reader for retrieval), and the grounding instructions in the system prompt are documented in the security and governance note for this project, and are referenced here rather than repeated.